Annotated Bibliography

Ian Chen

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Contents

Community Search

Bagrow: Evaluating local community methods in networks **bagrow2008evaluating**

James P Bagrow. "Evaluating local community methods in networks". In: *Journal of Statistical Mechanics: Theory and Experiment* 2008.05 (May 2008), P05001. ISSN: 1742-5468. DOI: `10.1088/1742-5468/2008/05/p05001`.

Annotations: **05/21/26**: I don't like this paper (old, modularity focused). Benchmark graphs: generate a BA($N = 512$, $m_0 = 8$) graph, partition nodes into ground truth communities, and rewire inter-community edges to form $Q \approx 1/2$. Accuracy via NMI. Their alleged take-home message is that stopping rule is more important than agglomeration scheme, and should be the focus of future work.

Abstract: We present a new benchmarking procedure that is unambiguous and specific to local community finding methods, allowing one to compare the accuracy of various methods. We apply this to new and existing algorithms. A simple class of synthetic benchmark networks is also developed, capable of testing properties specific to these local methods.

Barbieri et al.: Efficient and effective community search **barbieri2015efficient**

Nicola Barbieri et al. "Efficient and effective community search". In: *Data Mining and Knowledge Discovery* 29.5 (June 2015), pp. 1406–1433. ISSN: 1573-756X. DOI: `10.1007/s10618-015-0422-1`.

Annotations: **05/21/26**: k-core community search. Efficient means runtime; local search is desired. Effective means that we want to find small clusters. Seems to supercede Cui et al. [4]. Addressing efficiency, they propose ShellStruct (which finds maximal k-core). Addressing effectiveness, they propose a greedy heuristic that prunes k-core via a local search. They evaluate both single-node and multi-node queries.

Abstract: Community search is the problem of finding a good community for a given set of query vertices. One of the most studied formulations of community search asks for a connected subgraph that contains all query vertices and maximizes the minimum degree. All existing approaches to min-degree-based community search suffer from limitations concerning efficiency, as they need to visit (large part of) the whole input graph, as well as accuracy, as they output communities quite large and not really cohesive. Moreover, some existing methods lack generality: they handle only single-vertex queries, find communities that are not optimal in terms of minimum degree, and/or require input parameters. In this work we advance the state of the art on community search by proposing a novel method that overcomes all these limitations: it is in general more efficient and effective–one/two orders of magnitude on average, it can handle multiple query vertices, it yields optimal communities, and it is parameter-free. These properties are confirmed by an extensive experimental analysis performed on various real-world graphs.

Chu et al.: Finding the Best k in Core Decomposition: A Time and Space Optimal Solution **chu2020finding**

Deming Chu et al. "Finding the Best k in Core Decomposition: A Time and Space Optimal Solution". In: *2020 IEEE 36th International Conference on Data Engineering (ICDE)*. IEEE, Apr. 2020, pp. 685–696. DOI: `10.1109/icde48307.2020.00065`.

Abstract: The mode of k-core and its hierarchical decomposition have been applied in many areas, such as sociology, the world wide web, and biology. Algorithms on related studies often need an input value of parameter k, while there is no existing solution other than manual selection. In this paper, given a graph and a scoring metric, we aim to efficiently find the best value of k such that the score of the k-core (or k-core set) is the highest. The problem is challenging because there are various community scoring metrics and the computation is costly on large datasets. With the well-designed vertex ordering techniques, we propose time and space optimal algorithms to compute the best k, which are applicable to most community metrics. The proposed algorithms can compute the score of every k-core (set) and can benefit the solutions

to other k-core related problems. Extensive experiments are conducted on 10 real-world networks with size up to billion-scale, which validates both the efficiency of our algorithms and the effectiveness of the resulting k-cores.

Cui et al.: Local search of communities in large graphs — cui2014local

Wanyun Cui et al. "Local search of communities in large graphs". In: *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*. SIGMOD/PODS'14. ACM, June 2014, pp. 991–1002. DOI: 10.1145/2588555.2612179.

Annotations: **05/26/26**: First, they describe heuristics for smallest thresholded k-core commuity, and seem to do pretty good here. They run a BestFirstSearch around the query *node*, but early stop whenever we found a k-core, where Best has several heuristics. There exists a Best function that leads to optimal solutions (but is NP-Hard to compute). Their work on maximal k-core is a bit of sleight-of-hand foolery; to guarantee exact solutions, their only early-stopping rule is a sufficient condition for detecting when the node's coreness is equal to the degeneracy of a graph. They mostly test on DBLP, with $\langle k \rangle \approx 3$, m about a million. **05/21/26**: k-core community search, with minimum size, with single node queries, k as an argument. Finding small communities is good for runtime.

Abstract: Community search is important in social network analysis. For a given vertex in a graph, the goal is to find the best community the vertex belongs to. Intuitively, the best community for a given vertex should be in the vicinity of the vertex. However, existing solutions use global search to find the best community. These algorithms, although straight-forward, are very costly, as all vertices in the graph may need to be visited. In this paper, we propose a local search strategy, which searches in the neighborhood of a vertex to find the best community for the vertex. We show that, because the minimum degree measure used to evaluate the goodness of a community is not monotonic, designing effciient local search solutions is a very challenging task. We present theories and algorithms of local search to address this challenge. The efficiency of our local search strategy is verifeid by extensive experiments on both synthetic networks and a variety of real networks with millions of nodes.

Fang et al.: A survey of community search over big graphs — fang2019survey

Yixiang Fang et al. "A survey of community search over big graphs". In: *The VLDB Journal* 29.1 (July 2019), pp. 353–392. ISSN: 0949-877X. DOI: 10.1007/s00778-019-00556-x.

Annotations: **05/21/26**: This is the paper we always start from for community search, with a pretty comprehensive description of algorithms and problems, and definitions. Good as a reference.

Abstract: With the rapid development of information technologies, various big graphs are prevalent in many real applications (e.g., social media and knowledge bases). An important component of these graphs is the network community. Essentially, a community is a group of vertices which are densely connected internally. Community retrieval can be used in many real applications, such as event organization, friend recommendation, and so on. Consequently, how to efficiently find high-quality communities from big graphs is an important research topic in the era of big data. Recently, a large group of research works, called community search, have been proposed. They aim to provide effciient solutions for searching high-quality communities from large networks in real time. Nevertheless, these works focus on different types of graphs and formulate communities in different manners, and thus, it is desirable to have a comprehensive review of these works. In this survey, we conduct a thorough review of existing community search works. Moreover, we analyze and compare the quality of communities under their models, and the performance of different solutions. Furthermore, we point out new research directions. This survey does not only help researchers to have better understanding of existing community search solutions, but also provides practitioners a better judgment on choosing the proper solutions.

Gou et al.: A Comprehensive Survey and Experimental Study of Learning-Based Community Search gou2025comprehensive

Xiaoxuan Gou et al. "A Comprehensive Survey and Experimental Study of Learning-Based Community Search". In: *Proceedings of the VLDB Endowment* 18.9 (May 2025), pp. 2941–2954. ISSN: 2150-8097. DOI: 10.14778/3746405.3746419.

Abstract: Given a graph G and a query node q, the goal of community search (CS) is to find a structurally cohesive subgraph from G that contains q. Significant progress has been made in community search using deep learning in recent years. To the best of our knowledge, no existing work has provided a comprehensive review of learning-based community search methods. Additionally, we find that: (1) Existing methods offer diverse definitions or descriptions of communities, which require systematic summarization. (2) The methods rely on distinct metrics for limited community assessment. (3) Overhead evaluations of the methods vary and exhibit certain biases. Therefore, a comprehensive survey and experimental study are essential to achieve four key objectives: designing a unified pipeline, clarifying community definitions, enriching community evaluation, and establishing overhead assessment. In this paper, we first propose a unified pipeline for these methods, highlighting techniques. We categorize community definitions and analyze the relationships between identified communities. Beyond that, we proposed several community metrics to evaluate the communities comprehensively. Moreover, we introduce a more detailed overhead evaluation approach that considers resource consumption during both the training and search phases. Finally, we employ the proposed community evaluation metrics and overhead assessment framework to evaluate and analyze the methods, examine correlations among metrics, and explore the effects of several commonly used techniques.

Gou et al.: Effective and Efficient Community Search with Graph Embeddings gou2023effective

Xiaoxuan Gou et al. "Effective and Efficient Community Search with Graph Embeddings". In: *ECAI 2023*. IOS Press, Sept. 2023. ISBN: 9781643684376. DOI: 10.3233/faia230358.

Annotations: **05/21/26**: k-core based community search (query sets) via graph convolutional network; requires training; I hate machine learning papers. Sometimes okay, but also cases there get $<$ 50% F1 score with true solution. maximum dataset network size is 60,000.

Abstract: Given a graph G and a query node q , community search (CS) seeks a cohesive subgraph from G that contains q . CS has gained much research interests recently. In the database research community, researchers aim to find the most cohesive subgraph satisfying a specific community model (e.g., k -core or k -truss) via graph traversal. These works obtain good precision, however suffering from the low efficiency issue. In the AI research community, a new thought of using the deep learning model to support CS without relying on graph traversal emerges. Supervised end-to-end models using GCN are presented, which perform efficiently, but leave a large room for precision improvement. None of them can achieve a good balance between the efficiency and effectiveness. This motivates our solution: First, we present an offline community-injected graph embedding method to preserve the community's cohesiveness features into the learned node representations. Second, we resort to a proximity graph (PG) built from node representations, to quickly return the community online. Moreover, we develop a self-augmented method based on KL divergence to further optimize node representations. Extensive experiments on seven real-world graphs show our solution's superiority on effectiveness (at least 39.3% improvement) and efficiency (one to two orders of magnitude faster).

Huang et al.: Community Search over Big Graphs: Models, Algorithms, and Opportunities huang2017community

Xin Huang, Laks V.S. Lakshmanan, and Jianliang Xu. "Community Search over Big Graphs: Models, Algorithms, and Opportunities". In: *2017 IEEE 33rd International Conference on Data Engineering (ICDE)*. IEEE, Apr. 2017, pp. 1451–1454. DOI: 10.1109/icde.2017.211.

Annotations: **05/21/26**: Short list of applications and problems in community search. Treat this similar to Fang et al. [5].

Jiang et al.: GNN-Based Persistent K-core Community Search in Temporal Graphs jiang2024gnnbased

Zongli Jiang et al. "GNN-Based Persistent K-core Community Search in Temporal Graphs". In: *2024 IEEE International Conference on Big Data (BigData)*. IEEE, Dec. 2024, pp. 569–578. DOI: 10.1109/bigdata62323.2024.10825281.

Abstract: The goal of community search is to provide effective solutions for real-time, high-quality community searches within large networks. In many practical applications, such as event organization and friend recommendations, discovering various community structures within a network is crucial for users. However, existing community search algorithms rarely address issues within temporal graphs, and those that do often have two main limitations: (1) traditional community search methods become inefficient and experience significant increases in computation time when scaled to large graphs; (2) while GNN-based community search methods for temporal graphs offer generalizability, they often focus solely on community connectivity and lack cohesiveness. Therefore, we propose a new model PK-GCN, based on Graph Neural Networks (GNNs), to identify persistent k-core communities in temporal networks. This model can handle dynamic changes in temporal graphs and identify communities that persist over time. Compared to existing community search methods, our model not only finds communities with tighter structures but also allows for dynamic queries based on user input without needing retraining. Specifically, our model constructs features by integrating k-core information from core decomposition, graph features, and query features, resulting in more expressive node representations. Additionally, we designed a flexible dynamic query mechanism that allows users to input time information to query communities. Experiments on multiple datasets demonstrate that our model outperforms other GNN-based community search algorithms in F1-score.

Li et al.: Efficient progressive minimum k-core search li2020efficient

Conggai Li et al. "Efficient progressive minimum k-core search". In: *Proceedings of the VLDB Endowment* (2020). DOI: https://doi.org/10.14778/3368289.3368300.

Annotations: **05/21/26**: Instead of addressing the theoretically solved *maximum* k-core search, this paper focuses on approximating (as is NP-Hard) the smallest community that has the k-core property (for k an argument to the method). Previous works uses an expansion heuristic to add nodes. This paper describes a lot of theoretical structural results (that I have not read in detail), and describes an algorithm with an approximation ratio c (user-specified), and also for query sets. It does a lot better than previous methods, and scales to graphs with billions of edges in minutes (k=10) (so this is a different problem than the one in the survey Fang et al. [5].

Abstract: As one of the most representative cohesive subgraph models, k-core model has recently received significant attention in the literature. In this paper, we investigate the problem of the minimum k-core search: given a graph G, an integer k and a set of query vertices $Q = \{q\}$, we aim to find the smallest k-core subgraph containing every query vertex $q \in Q$. It has been shown that this problem is NP-hard with a huge search space, and it is very challenging to find the optimal solution. There are several heuristic algorithms for this problem, but they rely on simple scoring functions and there is no guarantee as to the size of the resulting subgraph, compared with the optimal solution. Our empirical study also indicates that the size of their resulting subgraphs may be large in practice. In this paper, we develop an effective and efficient progressive algorithm, namely PSA, to provide a good trade-of between the quality of the result and the search time. Novel lower and upper bound techniques for the minimum k-core search are designed. Our extensive experiments on 12 real-life graphs demonstrate the effectiveness and efficiency of the new techniques.

Li et al.: Influential community search in large networks li2015influential

Rong-Hua Li et al. “Influential community search in large networks”. In: *Proceedings of the VLDB Endowment* 8.5 (Jan. 2015), pp. 509–520. ISSN: 2150-8097. DOI: 10.14778/2735479.2735484.

Annotations: **05/21/26**: For graphs with node weights (typically centrality), this paper describes k-influence as the (not necessarily maximal) k-core subgraph that maximizes the minimum weight, and shows that greedy peeling is optimal. They then also describe ShellStruct (under a different name).

Abstract: Community search is a problem of finding densely connected subgraphs that satisfy the query conditions in a network, which has attracted much attention in recent years. However, all the previous studies on community search do not consider the influence of a community. In this paper, we introduce a novel community model called k-influential community based on the concept of k-core, which can capture the influence of a community. Based on the new community model, we propose a linear-time online search algorithm to find the top-r k-influential communities in a network. To further speed up the influential community search algorithm, we devise a linear-space index structure which supports efficient search of the top-r k-influential communities in optimal time. We also propose an efficient algorithm to maintain the index when the network is frequently updated. We conduct extensive experiments on 7 real-world large networks, and the results demonstrate the efficiency and effectiveness of the proposed methods.

Lin et al.: Effective and Efficient Conductance-Based Community Search at Billion Scale lin2025effective

Longlong Lin et al. “Effective and Efficient Conductance-Based Community Search at Billion Scale”. In: *IEEE Transactions on Big Data* 11.6 (Dec. 2025), pp. 3170–3184. ISSN: 2372-2096. DOI: 10.1109/tbdata.2025.3588028.

Annotations: **06/12/26**: The related work for expansion based community search is the Local from Sozio and Gionis [16] and Cui et al. [4], a related work for k-truss and modularity, and spectral (random-walk) methods (namely PPR-Nibble). They take the goodness function from the spectral methods (i.e. conductance), but enforce connectivity. Their first algorithm (PPRCS) is the same as PPR-Nibble, but restricted sweeps to connected subsets. They improve the quality of the algorithm in a 4-phase framework: 1. candidate subgraph (BFS + distance), 2. seed (maximal clique), 3. greedy expansion, and 4. greedy pruning (where greedy means optimizing single-step deltas in the objective function). They repeat 3 and 4 until convergence. On the SNAP networks with ground truth (biggest Friendster), they find 1. SCCS fastest, then PPRCS, 2. only SCCS and truss goodness find the small ≈ 100 node communities as in ground truth, 3. SCCS best F1-score (disregard this result as empirical networks). They provide software, and a nice comparison of very different goodness functions (albeit accuracy is suspect). **05/24/26**: SOTA for scaling community search. Have not read in detail.

Abstract: Community search is a widely studied semi-supervised graph clustering problem, retrieving a high-quality connected subgraph containing the user-specified query vertex. However, existing methods primarily focus on cohesiveness within the community but ignore the sparsity outside the community, obtaining sub-par results. Inspired by this, we adopt the well-known conductance metric to measure the quality of a community and introduce a novel problem of conductance-based community search (CCS). CCS aims at finding a subgraph with the smallest conductance among all connected subgraphs that contain the query vertex. We prove that the CCS problem is NP-hard. To efficiently query CCS, a four-stage subgraph-conductance-based community search algorithm, SCCS, is proposed. Specifically, we first greatly reduce the entire graph using local sampling techniques. Then, a three-stage local optimization strategy is employed to continuously refine the community quality. Namely, we first utilize a seeding strategy to obtain an initial community to enhance its internal cohesiveness. Then, we iteratively add qualified vertices in the expansion stage to guarantee the internal cohesiveness and external sparsity of the community. Finally, we gradually remove unqualified vertices during the verification stage. Extensive experiments on real-world datasets containing one billion-scale graph and synthetic datasets show the effectiveness, efficiency, and scalability of our solutions.

Lu et al.: On Time-optimal (k, p)-core Community Search in Dynamic Graphs lu2022timeoptimal

Zhao Lu et al. "On Time-optimal (k, p)-core Community Search in Dynamic Graphs". In: *2022 IEEE 38th International Conference on Data Engineering (ICDE)*. IEEE, May 2022, pp. 1396–1407. DOI: 10.1109/icde53745.2022.00109.

Annotations: **05/21/26**: Describes a new notion of (k, p)-Core, where it is a k-core such that p fraction of its neighbors in the subgraph. the maximal subgraph (solution to their proposed problem) may not be connected. They propose an index (using anchored union-find, similar to Barbieri et al. [2] that supports dynamic updates. time-optimal refers to queries, not the index construction.

Abstract: Community search aims to find cohesive subgraphs containing certain vertices, attracting increasing interest recently. However, existing cohesive models such as k-core mainly focus on the dense connections inside the community, and neglect the interactions with the vertices outside. In this paper, we study the (k,p) -core community search (KPCS) problem in dynamic graphs, i.e., find the maximal connected subgraph containing a query vertex where each vertex has at least k neighbors and at least p fraction of its neighbors in the subgraph. Such fraction and connectivity constraints bring non-trivial challenges to the online community search in dynamic graphs. Thus, we design a space-efficient $O(m)$ where m is the edge number) index KPForest which can support time-optimal (k,p) -core community search. We also propose novel construction and maintenance algorithms to record and update the (k,p) value and the connectivity information for dynamic graphs correctly and efficiently. Extensive experimental studies on ten real-world datasets show that our index can support community search with two orders of magnitude speedup at a small cost of construction and maintenance compared with the baseline algorithms.

O'Brien et al.: Locally Estimating Core Numbers obrien2014locally

Michael P. O'Brien and Blair D. Sullivan. "Locally Estimating Core Numbers". In: *2014 IEEE International Conference on Data Mining*. IEEE, Dec. 2014, pp. 460–469. DOI: 10.1109/icdm.2014.136.

Annotations: **05/18/26**: This paper develops very good theory for neighborhood-based estimation of the core number. Mathematically precise with proofs, but easy to read and good examples/figures. They show empirically that a constant number of steps is a very good approximation, and theoretically it converges to the true coreness.

Abstract: Graphs are a powerful way to model interactions and relationships in data from a wide variety of application domains. In this setting, entities represented by vertices at the 'center' of the graph are often more important than those associated with vertices on the 'fringes'. For example, central nodes tend to be more critical in the spread of information or disease and play an important role in clustering/community formation. Identifying such 'core' vertices has recently received additional attention in the context of network experiments, which analyze the response when a random subset of vertices are exposed to a treatment (e.g. Inoculation, free product samples, etc). Specifically, the likelihood of having many central vertices in any exposure subset can have a significant impact on the experiment. We focus on using k-cores and core numbers to measure the extent to which a vertex is central in a graph. Existing algorithms for computing the core number of a vertex require the entire graph as input, an unrealistic scenario in many real world applications. Moreover, in the context of network experiments, the sub graph induced by the treated vertices is only known in a probabilistic sense. We introduce a new method for estimating the core number based only on the properties of the graph within a region of radius δ around the vertex, and prove an asymptotic error bound of our estimator on random graphs. Further, we empirically validate the accuracy of our estimator for small values of δ on a representative corpus of real data sets. Finally, we evaluate the impact of improved local estimation on an open problem in network experimentation posed by Ugander et al.

Sarıyüce et al.: Incremental k-core decomposition: algorithms and evaluation **sariyuce2016incremental**

Ahmet Erdem Sarıyüce et al. "Incremental k-core decomposition: algorithms and evaluation". In: *The VLDB Journal* 25.3 (Feb. 2016), pp. 425–447. ISSN: 0949-877X. DOI: `10.1007/s00778-016-0423-8`.

Annotations: **05/18/26**: This paper develops local traversal style algorithms that exactly update the coreness after an edge insertion/deletion in linear time. It does a lot of good pruning, but makes no theoretical progress, as in the worst case, all core numbers can change (e.g. adding an edge to turn path to cycle).

Abstract: A k-core of a graph is a maximal connected subgraph in which every vertex is connected to at least k vertices in the subgraph. k-core decomposition is often used in large-scale network analysis, such as community detection, protein function prediction, visualization, and solving NP-hard problems on real networks efficiently, like maximal clique finding. In many real-world applications, networks change over time. As a result, it is essential to develop efficient incremental algorithms for dynamic graph data. In this paper, we propose a suite of incremental k-core decomposition algorithms for dynamic graph data. These algorithms locate a small subgraph that is guaranteed to contain the list of vertices whose maximum k-core values have changed and efficiently process this subgraph to update the k-core decomposition. We present incremental algorithms for both insertion and deletion operations, and propose auxiliary vertex state maintenance techniques that can further accelerate these operations. Our results show a significant reduction in runtime compared to non-incremental alternatives. We illustrate the efficiency of our algorithms on different types of real and synthetic graphs, at varying scales. For a graph of 16 million vertices, we observe relative throughputs reaching a million times, relative to the non-incremental algorithms.

Sozio et al.: The community-search problem and how to plan a successful cocktail party **sozio2010communitysearch**

Mauro Sozio and Aristides Gionis. "The community-search problem and how to plan a successful cocktail party". In: *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*. KDD '10. ACM, July 2010, pp. 939–948. DOI: `10.1145/1835804.1835923`.

Annotations: **05/21/26**: This is the original paper for community search. The problem is to find the maximium score, connected (not necessarily the largest size community). They also often add a distance constraint. Describes the Local and Global search problem, and an exact greedy algorithm that optimizes a small family of objectives.

Abstract: A lot of research in graph mining has been devoted in the dis-covery of communities. Most of the work has focused in thescenario where communities need to be discovered with onlyreference to the input graph. However, for many interest-ing applications one is interested in finding the communityformed by a given set of nodes. In this paper we study aquery-dependent variant of the community-detection prob-lem, which we call the community-search problem: given agraph G, and a set of query nodes in the graph, we seek to find a subgraph of G that contains the query nodes and itis densely connected.We motivate a measure of density based on minimum de-gree and distance constraints, and we develop an optimumgreedy algorithm for this measure. We proceed by charac-terizing a class of monotone constraints and we generalizeour algorithm to compute optimum solutions satisfying anyset of monotone constraints. Finally we modify the greedyalgorithm and we present two heuristic algorithms that find communities of size no greater than a specified upper bound.Our experimental evaluation on real datasets demonstratesthe efficiency of the proposed algorithms and the quality ofthe solutions we obtain.

Sun et al.: Index-Based Intimate-Core Community Search in Large Weighted Graphs **sun2022indexbased**

Longxu Sun et al. "Index-Based Intimate-Core Community Search in Large Weighted Graphs". In: *IEEE Transactions on Knowledge and Data Engineering* 34.9 (Sept. 2022), pp. 4313–4327. ISSN: 2326-3865. DOI: `10.1109/tkde.2020.3040762`.

Abstract: Community search that finds query-dependent communities has been studied on various kinds of graphs. As one instance of community search, intimate-core group (community) search over a weighted graph is to find a connected k-core containing all query nodes with the smallest group weight. However, existing state-of-the-art methods start from the maximal k-core to refine an answer, which is practically inefficient for large networks. In this paper, we develop an efficient framework, called local exploration k-core search (LEKS), to find intimate-core groups in graphs. We propose a small-weighted spanning tree to connect query nodes, and then expand the tree level by level to a connected k-core, which is finally refined as an intimate-core group. In addition, to support the intimate group search over large weighted graphs, we develop a weighted-core index (WC-index) and two new WC-index-based algorithms for expansion and refinement phases in LEKS. Specifically, we propose a WC-index-based expansion to efficiently find a candidate graph of intimate-core group, leveraging on a two-level expansion of k-breadth and 1-depth. We propose two graph removal strategies: coarse-grained refinement is designed for large graphs to delete a batch of nodes in a few iterations; fine-grained refinement is designed for small graphs to remove nodes carefully and achieve high-quality answers. Extensive experiments on real-life networks with ground-truth communities validate the effectiveness and efficiency of our proposed methods.

Tang et al.: Reliable community search in dynamic networks — tang2022reliable

Yifu Tang et al. "Reliable community search in dynamic networks". In: *Proceedings of the VLDB Endowment* 15.11 (July 2022), pp. 2826–2838. ISSN: 2150-8097. DOI: 10.14778/3551793.3551834.

Abstract: Searching for local communities is an important research problem that supports advanced data analysis in various complex networks, such as social networks, collaboration networks, cellular networks, etc. The evolution of such networks over time has motivated several recent studies to identify local communities in dynamic networks. However, these studies only utilize the aggregation of disjoint structural information to measure the quality and ignore the reliability of the communities in a continuous time interval. To fill this research gap, we propose a novel (θ, k)-core reliable community (CRC) model in the weighted dynamic networks, and define the problem of most reliable community search that couples the desirable properties of connection strength, cohesive structure continuity, and the maximal member engagement. To solve this problem, we first develop a novel edge filtering based online CRC search algorithm that can effectively filter out the trivial edge information from the networks while searching for a reliable community. Further, we propose an index structure, Weighted Core Forest-Index (WCF-index), and devise an index-based dynamic programming CRC search algorithm, that can prune a large number of insignificant intermediate results and support efficient query processing. Finally, we conduct extensive experiments systematically to demonstrate the efficiency and effectiveness of our proposed algorithms on eight real datasets under various experimental settings.

Wen et al.: I/O Efficient Core Graph Decomposition: Application to Degeneracy Ordering — wen2019io

Dong Wen et al. "I/O Efficient Core Graph Decomposition: Application to Degeneracy Ordering". In: *IEEE Transactions on Knowledge and Data Engineering* 31.1 (Jan. 2019), pp. 75–90. ISSN: 2326-3865. DOI: 10.1109/tkde.2018.2833070.

Annotations: **05/19/26**: Propagates an upper-bound for the coreness, a function of just a node's neighbors. Hence, with graph partitioning, we can scale k-core estimation to just use $O(n)$ space. The notation and writing is a bit clunky, but is overall pretty good work.

Abstract: Core decomposition is a fundamental graph problem with a large number of applications. Most existing approaches for core decomposition assume that the graph is kept in memory of a machine. Nevertheless, many real-world graphs are too big to reside in memory. In this paper, we study I/O efficient core decomposition following a semi-external model, which only allows node information to be loaded in memory. We propose a semi-external algorithm and an optimized algorithm for I/O efficient core decomposition. To handle dynamic graph updates, we firstly show that our algorithm can be naturally extended to handle

edge deletion. Then, we propose an I/O efficient core maintenance algorithm to handle edge insertion, and an improved algorithm to further reduce I/O and CPU cost. In addition, based on our core decomposition algorithms, we further propose an I/O efficient semi-external algorithm for degeneracy ordering, which is an important graph problem that is highly related to core decomposition. We also consider how to maintain the degeneracy order. We conduct extensive experiments on 12 real large graphs. Our optimal core decomposition algorithm significantly outperforms the existing I/O efficient algorithm in terms of both processing time and memory consumption. They are very scalable to handle web-scale graphs. As an example, we are the first to handle a web graph with 978.5 million nodes and 42.6 billion edges using less than 4.2 GB memory. We also show that our proposed algorithms for degeneracy order computation and maintenance can handle big graphs efficiently with small memory overhead.

Wu et al.: Robust Local Community Detection: On Free Rider Effect and Its Elimination. wu2015robust

Yubao Wu et al. "Robust Local Community Detection: On Free Rider Effect and Its Elimination." In: *Proc. VLDB Endow.* 8.7 (2015), pp. 798–809.

Annotations: **06/12/26**: Similar style to Traag, Van Dooren, and Nesterov [31]; theory about designing optimization functions, with a somewhat strange notion that leads to narrow scope of goodness. The formulation of free-rider is not super convincing to me (about sub/super-modularity), but is saying that adding sets that optimize the goodness function independently of the query should not make a better quality community. Seems fine to use if it was adopted like CPM; likely a similar gap between theory and empirical performance.

Abstract: Given a large network, local community detection aims at finding the community that contains a set of query nodes and also maximizes (minimizes) a goodness metric. This problem has recently drawn intense research interest. Various goodness metrics have been proposed. However, most existing metrics tend to include irrelevant subgraphs in the detected local community. We refer to such irrelevant subgraphs as free riders. We systematically study the existing goodness metrics and provide theoretical explanations on why they may cause the free rider effect. We further develop a query biased node weighting scheme to reduce the free rider effect. In particular, each node is weighted by its proximity to the query node. We define a query biased density metric to integrate the edge and node weights. The query biased densest subgraph, which has the largest query biased density, will shift to the neighborhood of the query nodes after node weighting. We then formulate the query biased densest connected subgraph (QDC) problem, study its complexity, and provide efficient algorithms to solve it. We perform extensive experiments on a variety of real and synthetic networks to evaluate the effectiveness and efficiency of the proposed methods.

Zhang et al.: Efficient community discovery with user engagement and similarity zhang2019efficient

Fan Zhang et al. "Efficient community discovery with user engagement and similarity". In: *The VLDB Journal* 28.6 (Oct. 2019), pp. 987–1012. ISSN: 0949-877X. DOI: 10.1007/s00778-019-00579-4.

Annotations: **05/21/26**: Propose a new objective, (k, r)-core, where k is the coreness and r is a threshold that all nodes within the community must have pairwise similarity at least r. NP-Hard. Lots of heuristics. Lots of good examples, too, but very long paper, although mostly well-written (although I am not interested).

Abstract: In this paper, we investigate the problem of (k,r)-core which intends to find cohesive subgraphs on social networks considering both user engagement and similarity perspectives. In particular, we adopt the popular concept of k-core to guarantee the engagement of the users (vertices) in a group (subgraph) where each vertex in a (k,r)-core connects to at least k other vertices. Meanwhile, we consider the pairwise similarity among users based on their attributes. Efficient algorithms are proposed to enumerate all maximal (k,r)-cores and find the maximum (k,r)-core, where both problems are shown to be NP-hard. Effective pruning

techniques substantially reduce the search space of two algorithms. A novel (k,k')-core based (k,r)-core size upper bound enhances the performance of the maximum (k,r)-core computation. We also devise effective search orders for two algorithms with different search priorities for vertices. Besides, we study the diversified (k,r)-core search problem to find l maximal (k,r)-cores which cover the most vertices in total. These maximal (k,r)-cores are distinctive and informationally rich. An efficient algorithm is proposed with a guaranteed approximation ratio. We design a tight upper bound to prune unpromising partial (k,r)-cores. A new search order is designed to speed up the search. Initial candidates with large size are generated to further enhance the pruning power. Comprehensive experiments on real-life data demonstrate that the maximal (k,r)-cores enable us to find interesting cohesive subgraphs, and performance of three mining algorithms is effectively improved by all the proposed techniques.

Community Detection

Fortunato et al.: Resolution limit in community detection **fortunato2007resolution**

Santo Fortunato and Marc Barthélemy. "Resolution limit in community detection". In: *Proceedings of the National Academy of Sciences* 104.1 (Jan. 2007), pp. 36–41. ISSN: 1091-6490. DOI: `10.1073/pnas.0605965104`.

Annotations: **05/18/26**: This paper addresses the $\sqrt{E}$ volume bias of modularity optimization. For two communities of volume $< \sqrt{2E}$, then if they are connected by even one edge, modularity is strictly increased after merging, and we conclude a limit of $\sqrt{2E}$ communities in the optimal modularity solution. This is a short, well-written paper with instructive examples, e.g. ring-of-cliques.

Abstract: Detecting community structure is fundamental for uncovering the links between structure and function in complex networks and for practical applications in many disciplines such as biology and sociology. A popular method now widely used relies on the optimization of a quantity called modularity, which is a quality index for a partition of a network into communities. We find that modularity optimization may fail to identify modules smaller than a scale which depends on the total size of the network and on the degree of interconnectedness of the modules, even in cases where modules are unambiguously defnied. This finding is confirmed through several examples, both in artificial and in real social, biological, and technological networks, where we show that modularity optimization indeed does not resolve a large number of modules. A check of the modules obtained through modularity optimization is thus necessary, and we provide here key elements for the assessment of the reliability of this community detection method.

Giatsidis et al.: CoreCluster: A Degeneracy Based Graph Clustering Framework **giatsidis2014corecluster**

Christos Giatsidis et al. "CoreCluster: A Degeneracy Based Graph Clustering Framework". In: *Proceedings of the AAAI Conference on Artificial Intelligence* 28.1 (June 2014). ISSN: 2159-5399. DOI: `10.1609/aaai.v28i1.8731`.

Annotations: **05/30/26**: Compare to METIS, where spectral clustering is a bit more expensive. Good connection in theorem 1, and also with the IKC paper about center-periphery and coreness. Short and sweet, but more evaluation would be desirable.

Abstract: Graph clustering or community detection constitutes an important task forinvestigating the internal structure of graphs, with a plethora of applications in several domains. Traditional tools for graph clustering, such asspectral methods, typically suffer from high time and space complexity. In thisarticle, we present CoreCluster, an efficient graph clusteringframework based on the concept of graph degeneracy, that can be used along withany known graph clustering algorithm. Our approach capitalizes on processing thegraph in a hierarchical manner provided by its core expansion sequence, anordered partition of the graph into different levels according to the k-coredecomposition. Such a partition provides a way to process the graph inan incremental manner that preserves its clustering structure, whilemaking the execution of the chosen clustering algorithm much faster due to thesmaller size of the graph's partitions onto which the algorithm operates.

Jones et al.: Community Cores: Removing Size Bias from Community Detection **jones2021community**

Isaac Jones et al. "Community Cores: Removing Size Bias from Community Detection". In: *Proceedings of the International AAAI Conference on Web and Social Media* 10.1 (Aug. 2021), pp. 603–606. ISSN: 2162-3449. DOI: `10.1609/icwsm.v10i1.14780`.

Annotations: **05/22/26**: This paper discusses an edge-reweighting scheme that addresses the resolution limit, given by $C-$ edge_betweenness. Then, they run Infomap, label-propagation, and Louvain on LFR networks, computing accuracy via generalized NMI (with overlapping communities). Using NMI is okay-ish since they want smaller communities. In general, results are not very strong, but does seem to often be beneficial (though random reweighting seems to do more-or-less the same thing)? Short paper; FastEnsemble [30] is probably better.

Abstract: Community discovery in social networks has received a significant amount of attention in the social media research community. The techniques developed by the community have become quite adept at identifying the large communities in a network, but often neglect smaller communities. Evaluation techniques also show this bias, as the resolution limit problem in modularity indicates. Small communities, however, account for a higher proportion of a social network's community membership and reveal important information about the members of these communities. In this work, we introduce a re-weighting method to improve both the over all performance of community detection algorithms and performance on small community detection.

Kannan et al.: On clusterings: Good, bad and spectral — kannan2004clusterings

Ravi Kannan, Santosh Vempala, and Adrian Vetta. "On clusterings: Good, bad and spectral". In: *Journal of the ACM* 51.3 (May 2004), pp. 497–515. ISSN: 1557-735X. DOI: 10.1145/990308.990313.

Annotations: **05/24/26**: This is an old paper analyzing the approximation guarantees of two spectral algorithms, based on min-conductance and ratio-cut respectively. Besides the theory, they have a pretty insightful discussion of the challenges in designing a good objective function, namely balancing disparities in community sizes (leading to normalizing mincut by volume), and the tradeoffs between pessimistic bounds (e.g. minimum cluster score) and outliers/node coverage. The authors provide instructive examples for the failure of minimum diameter, k-median, and even minimum conductance. Finally, a bi-criteria objective (α, ϵ) is proposed, requiring minimum conductance at least α and ϵ proportion of inter-community edges. Good link between theory and practice; foundational work but im a bit unsure of how ϵ deals with the outliers challenge.

Abstract: We motivate and develop a natural bicriteria measure for assessing the quality of a clustering that avoids the drawbacks of existing measures. A simple recursive heuristic is shown to have polylogarithmic worst-case guarantees under the new measure. The main result of the article is the analysis of a popular spectral algorithm. One variant of spectral clustering turns out to have effective worst-case guarantees; another finds a "good" clustering, if one exists.

Ovelgonne et al.: An ensemble learning strategy for graph clustering — ovelgonne2013ensemble

Michael Ovelgonne and Andreas Geyer-Schulz. "An ensemble learning strategy for graph clustering". In: *Graph Partitioning and Graph Clustering* (2013), pp. 187–205. ISSN: 1098-3627. DOI: 10.1090/conm/588/11701.

Annotations: **06/10/26**: This paper proposes CGGC (Core Groups Graph Cluster), a new ensemble method for clustering. Using a set of weak learners, they find core groups, i.e. vertices that are assigned to co-clustered in all clusterings, cluster a quotiented graph by this relation, and project it backwards (more-or-less another way of consensus). The authors then summarize several pipelines with their scheme. Because they only evaluate modularity, I don't really care for their results, but they do mention that label prop is worse as a modularity optimizer than their alternatives.

Abstract: This paper is on a graph clustering scheme inspired by ensemble learning. In short, the idea of ensemble learning is to learn several weak classifiers and use these weak classifiers to determine a strong classifier. In this contribution, we use the generic procedure of ensemble learning and determine several weak graph clusterings (with respect to the objective function). From the partition given by the maximal overlap of these clusterings (the cluster cores), we continue the search for a strong clustering. We demonstrate the performance of this scheme by using it to maximize the modularity of a graph clustering. We show, that the quality of the initial weak clusterings is of minor importance for the quality of the final result of the scheme if we iterate the process of restarting from maximal overlaps.

Park et al.: Well-connectedness and community detection **park2024wellconnectedness**

Minhyuk Park et al. "Well-connectedness and community detection". In: *PLOS Complex Systems* 1.3 (Nov. 2024). Ed. by Réka Albert, e0000009. ISSN: 2837-8830. DOI: `10.1371/journal.pcsy.0000009`.

Annotations: **05/22/26**: Recall the characterization of communities as dense, cohesive, and separated subsets of the network. Recall the findings of Fortunato and Barthélemy [22], which recommend reclustering each module recursively. This paper evaluates a meta-clustering algorithm (CM) that, given a base clustering algorithm, recursively reclusters the modules using the base clustering algorithm, repeating until the stopping rule of "well-connectedness", and minimum size. Here, well-connectedness means that the minimum cut of a module of size n is at least $f(n)$, where they recommend $\log_b(n)$ for its stricter requirement for small module than on large modules. They evaluate statistics on some empirical networks, mostly arguing that while they reduce the quality of the objective, the guarantee of connectivity is important. They also evaluate accuracy on synthetic networks (LFR, ER), but I don't quite understand this part yet.

Abstract: Community detection methods help reveal the meso-scale structure of complex networks. Integral to detecting communities is the expectation that communities in a network are edge-dense and "well-connected". Surprisingly, we find that five different community detection methods-the Leiden algorithm optimizing the Constant Potts Model, the Leiden algorithm optimizing modularity, Infomap, Markov Cluster (MCL), and Iterative k-core (IKC)-identify communities that fail even a mild requirement for well-connectedness. To address this issue, we have developed the Connectivity Modifier (CM), which iteratively removes small edge cuts and re-clusters until communities are well-connected according to a user-specified criterion. We tested CM on real-world networks ranging in size from approximately 35,000 to 75,000,000 nodes. Post-processing of the output of community detection methods by CM resulted in a reduction in node coverage. Results on synthetic networks show that the CM algorithm generally maintains or improves accuracy in recovering true communities. This study underscores the importance of network clusterability-the fraction of a network that exhibits community structure-and the need for more models of community structure where networks contain nodes that are not assigned to communities. In summary, we address well-connectedness as an important aspect of clustering and present a scalable open-source tool for well-connected clusters.

Raghavan et al.: Near linear time algorithm to detect community structures in large-scale networks **raghavan2007near**

Usha Nandini Raghavan, Reka Albert, and Soundar Kumara. "Near linear time algorithm to detect community structures in large-scale networks". In: *Physical Review E* 76.3 (Sept. 2007). ISSN: 1550-2376. DOI: `10.1103/physreve.76.036106`.

Annotations: **06/10/26**: This paper proposes Label Detection, a clustering algorithm that under convergence, guarantees each community is strong, i.e. each node has (not strictly) more neighbors inside the cluster than outside. Starting from a singleton cluster, they iterate the each node (permutationally) and move it to the majority of the neighbors. Pralat, Kaminski, and Theberge's book mentions that this may not converge in a finite number of iterations. This algorithm has high variability and may find many different clusterings, so is recommended to run multiple times. They also show that by using labels that are cartesian products of multiple weak learners, Label Propagation can be used as an ensemble method. As with old papers, their testing is extremely limited (Zachary karate club and US football), so its empirical characteristics are somewhat unknown.

Abstract: Community detection and analysis is an important methodology for understanding the organization of various real-world networks and has applications in problems as diverse as consensus formation in social communities or the identification of functional modules in biochemical networks. Currently used algorithms that identify the community structures in large-scale real-world networks require a priori information such as the number and sizes of communities or are computationally expensive. In this paper we investigate a simple label propagation algorithm that uses the network structure alone as its guide and requires neither optimization of a predefined objective function nor prior information about the communities. In our algorithm

every node is initialized with a unique label and at every step each node adopts the label that most of its neighbors currently have. In this iterative process densely connected groups of nodes form a consensus on a unique label to form communities. We validate the algorithm by applying it to networks whose community structures are known. We also demonstrate that the algorithm takes an almost linear time and hence it is computationally less expensive than what was possible so far.

Rosvall et al.: Maps of random walks on complex networks reveal community structure rosvall2008maps

Martin Rosvall and Carl T. Bergstrom. "Maps of random walks on complex networks reveal community structure". In: *Proceedings of the National Academy of Sciences* 105.4 (Jan. 2008), pp. 1118–1123. ISSN: 1091-6490. DOI: `10.1073/pnas.0706851105`.

Annotations: **05/18/26**: Very interesting paper with a distinctive approach to clustering. It defines what it wants to capture (information flow), substitutes a reasonable surrogate (probability flow), and very elegantly describes an optimization function (description length via multi-layer encoding) to capture it. Lots of very instructive diagrams, and overall very nicely written.

Abstract: To comprehend the multipartite organization of large-scale biological and social systems, we introduce an information theoretic approach that reveals community structure in weighted and directed networks. We use the probability flow of random walks on a network as a proxy for information flows in the real system and decompose the network into modules by compressing a description of the probability flow. The result is a map that both simplifies and highlights the regularities in the structure and their relationships. We illustrate the method by making a map of scientific communication as captured in the citation patterns of >6,000 journals. We discover a multicentric organization with fields that vary dramatically in size and degree of integration into the network of science. Along the backbone of the network–including physics, chemistry, molecular biology, and medicine–information flows bidirectionally, but the map reveals a directional pattern of citation from the applied fields to the basic sciences.

Tabatabaee et al.: FastEnsemble: Scalable ensemble clustering on large networks tabatabaee2025fastensemble

Yasamin Tabatabaee et al. "FastEnsemble: Scalable ensemble clustering on large networks". In: *PLOS Complex Systems* 2.10 (Oct. 2025). Ed. by Tobias Braun, e0000069. ISSN: 2837-8830. DOI: `10.1371/journal.pcsy.0000069`.

Abstract: Many community detection algorithms are inherently stochastic, leading to variations in their output depending on input parameters and random seeds. This variability makes the results of a single run of these algorithms less reliable. Moreover, different clustering algorithms, optimization criteria (e.g., modularity and the Constant Potts model), and resolution values can result in substantially different partitions on the same network. Consensus clustering methods, such as Ensemble Clustering for Graphs (ECG) and FastConsensus, have been proposed to reduce the instability of non-deterministic algorithms and improve their accuracy by combining a set of partitions resulting from multiple runs of a clustering algorithm. In Complex Networks and their Applications 2024, we introduced FastEnsemble, a new consensus clustering method; here we present a more extensive evaluation of this method. Our results on both real-world and synthetic networks show that FastEnsemble produces more accurate clusterings than two other consensus clustering methods, ECG and FastConsensus, for many model conditions. Furthermore, FastEnsemble is fast enough to be used on networks with more than 3 million nodes, and so improves on the speed and scalability of FastConsensus. Finally, we showcase the utility of consensus clustering methods in mitigating the effect of resolution limit and clustering networks that are only partially covered by communities.

Traag et al.: Narrow scope for resolution-limit-free community detection traag2011narrow

V. A. Traag, P. Van Dooren, and Y. Nesterov. "Narrow scope for resolution-limit-free community detection". In: *Physical Review E* 84.1 (July 2011). ISSN: 1550-2376. DOI: `10.1103/physreve.84.016114`.

Annotations: **05/19/26**: If a resolution-limit free algorithm produces a clustering $\mathcal{C} = \{C_i\}$, then it should produce the same clustering on any induced subgraph of a union of clusters. Traag et al. describes *clustering* quality objectives, and show that *only* their proposed objective CPM, or something very similar to it, can be resolution limit free. Finally, they perform experiments showing good empirical performance. This paper seems OK; I don't like the problem of having to choose the right resolutin parameter. Good theory, and much much better than modularity. Some more examples and intuition behind CPM would be good, but is provided in the Leiden suppl.

Abstract: Detecting communities in large networks has drawn much attention over the years. While modularity remains one of the more popular methods of community detection, the so-called resolution limit remains a significant drawback. To overcome this issue, it was recently suggested that instead of comparing the network to a random null model, as is done in modularity, it should be compared to a constant factor. However, it is unclear what is meant exactly by "resolution-limit-free," that is, not suffering from the resolution limit. Furthermore, the question remains what other methods could be classified as resolution-limit-free. In this paper we suggest a rigorous definition and derive some basic properties of resolution-limit-free methods. More importantly, we are able to prove exactly which class of community detection methods are resolution-limit-free. Furthermore, we analyze which methods are not resolution-limit-free, suggesting there is only a limited scope for resolution-limit-free community detection methods. Finally, we provide such a natural formulation, and show it performs superbly.

Wedell et al.: Center-periphery structure in research communities wedell2022centerperiphery

Eleanor Wedell et al. "Center-periphery structure in research communities". In: *Quantitative Science Studies* 3.1 (2022), pp. 289–314. ISSN: 2641-3337. DOI: `10.1162/qss_a_00184`.

Annotations: **05/30/26**: Quite a nicely written paper, and describes iterative k-core clustering (IKC). Some theory and lots of experiments. A *kmp* core-periphery has every core with coreness k and positive modularity, with periphery with p connections to core nodes. They find *km*-valid by IKC (k-core extraction), refine with Graclus (seems similar to CM or WCC[27]), assemble periphery by moving outliers to clusters (i.e. label-propagation). A final step ensures that Graclus refinement maintains m-validness. Interesting relationship with k-core and sign of modularity: $k = 1$ is +ve, $2 - 39$ is $-$ve, and 40+ is +ve.. They analyze CEN using George's knowledge of marker nodes; pretty embedding and seems to produce reasonable outputs, arguing that center-periphery structure exists in CEN. IKC is very different from Leiden, and inspiration should be taken in their analysis.

Abstract: Abstract Clustering and community detection in networks are of broad interest and have been the subject of extensive research that spans several fields. We are interested in the relatively narrow question of detecting communities of scientific publications that are linked by citations. These publication communities can be used to identify scientists with shared interests who form communities of researchers. Building on the well-known k-core algorithm, we have developed a modular pipeline to find publication communities with center-periphery structure. Using a quantitative and qualitative approach, we evaluate community finding results on a citation network consisting of over 14 million publications relevant to the field of extracellular vesicles. We compare our approach to communities discovered by the widely used Leiden algorithm for community finding.

Community Evaluation

Jerdee et al.: Normalized mutual information is a biased measure for classification and community detection **jerdee2025normalized**

Maximilian Jerdee, Alec Kirkley, and Mark Newman. "Normalized mutual information is a biased measure for classification and community detection". In: *Nature Communications* 16.1 (Dec. 2025). ISSN: 2041-1723. DOI: 10.1038/s41467-025-66150-8.

Annotations: **05/24/26**: This is yet another way of improving NMI (e.g. AMI and RMI). I don't think they compare it to AMI or RMI. NMI is size-biased (prefers many estimated groups), and the normalization factor should be different for different clusterings (asymmetric). Working through the math may be instructive but not easy. They only compare to Infomap and modularity optimization, neither are known for their accuracy.

Abstract: Normalized mutual information is widely used as a similarity measure for evaluating the performance of clustering and classification algorithms. In this paper, we argue that results returned by the normalized mutual information are biased for two reasons: first, because they ignore the information content of the contingency table and, second, because their symmetric normalization introduces spurious dependence on algorithm output. We introduce a modified version of the mutual information that remedies both of these shortcomings. As a practical demonstration of the importance of using an unbiased measure, we perform extensive numerical tests on a basket of popular algorithms for network community detection and show that one's conclusions about which algorithm is best are significantly affected by the biases in the traditional mutual information. From blood tests to friend groups, normalized mutual information is widely used to assess similarity between classifications, outcomes, or labelings of data. Here the authors demonstrate systematic biases of this measure and propose a modification that eliminates them.

Kannan et al.: On clusterings: Good, bad and spectral **kannan2004clusterings**

Ravi Kannan, Santosh Vempala, and Adrian Vetta. "On clusterings: Good, bad and spectral". In: *Journal of the ACM* 51.3 (May 2004), pp. 497–515. ISSN: 1557-735X. DOI: 10.1145/990308.990313.

Annotations: **05/24/26**: This is an old paper analyzing the approximation guarantees of two spectral algorithms, based on min-conductance and ratio-cut respectively. Besides the theory, they have a pretty insightful discussion of the challenges in designing a good objective function, namely balancing disparities in community sizes (leading to normalizing mincut by volume), and the tradeoffs between pessimistic bounds (e.g. minimum cluster score) and outliers/node coverage. The authors provide instructive examples for the failure of minimum diameter, k-median, and even minimum conductance. Finally, a bi-criteria objective (α, ϵ) is proposed, requiring minimum conductance at least α and ϵ proportion of inter-community edges. Good link between theory and practice; foundational work but im a bit unsure of how ϵ deals with the outliers challenge.

Abstract: We motivate and develop a natural bicriteria measure for assessing the quality of a clustering that avoids the drawbacks of existing measures. A simple recursive heuristic is shown to have polylogarithmic worst-case guarantees under the new measure. The main result of the article is the analysis of a popular spectral algorithm. One variant of spectral clustering turns out to have effective worst-case guarantees; another finds a "good" clustering, if one exists.

Vu-Le et al.: EC-SBM synthetic network generator **vu-le2025ecsbm**

The-Anh Vu-Le et al. "EC-SBM synthetic network generator". In: *Applied Network Science* 10.1 (May 2025). ISSN: 2364-8228. DOI: 10.1007/s41109-025-00701-2.

Annotations: **05/19/26**: A high quality network generator should have similar statistics to empirical networks, and be able to scale to large networks. In particular, they want to recover community structure, such as clustering coefficient and connectivity of communities. To ensure connectivity, they overlay a k-edge connected spanning subnetwork with an SBM network, with final step of degree correction, finding the best parameters from a SBM+WCC clustering. They perform the best on all metrics (compare to RECCS, SBM, ...) except connectivity, with RECCS being a bit better. This paper is quite long and has a comprehensive set of experiments; I wonder what the network community profile looks like. Seems okay to use as a blackbox and not think about its mechanics.

Abstract: Generating high-quality synthetic networks with realistic community structure is vital to effectively evaluate community detection algorithms. In this study, we propose a new synthetic network generator called the Edge-Connected Stochastic Block Model (EC-SBM). The goal of EC-SBM is to take a given clustered real-world network and produce a synthetic network that resembles the clustered real-world network with respect to both network and community-specific criteria. In particular, we focus on simulating the internal edge connectivity of the clusters in the reference clustered network. Our performance study on large real-world networks shows that EC-SBM is generally more accurate with respect to network and community criteria than currently used approaches for this problem. Furthermore, we demonstrate that EC-SBM can complete analyses on several real-world networks with millions of nodes.

Leskovec et al.: Graph evolution: Densification and shrinking diameters leskovec2007graph

Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. "Graph evolution: Densification and shrinking diameters". In: *ACM Transactions on Knowledge Discovery from Data* 1.1 (Mar. 2007), p. 2. ISSN: 1556-472X. DOI: 10.1145/1217299.1217301.

Annotations: **05/30/26**: This is the completed study of Leskovec, Kleinberg, and Faloutsos [36]. It is somewhat hard to tell what they add though, besides more discussion and more-or-less same experiments. They play around with more parameters of their model, such as multiple ambassadors in the forest-fire model, and evaluate the fit to real-world networks more closely.

Abstract: How do real graphs evolve over time? What are normal growth patterns in social, technological, and information networks? Many studies have discovered patterns in static graphs, identifying properties in a single snapshot of a large network or in a very small number of snapshots; these include heavy tails for in- and out-degree distributions, communities, small-world phenomena, and others. However, given the lack of information about network evolution over long periods, it has been hard to convert these findings into statements about trends over time. Here we study a wide range of real graphs, and we observe some surprising phenomena. First, most of these graphs densify over time with the number of edges growing superlinearly in the number of nodes. Second, the average distance between nodes often shrinks over time in contrast to the conventional wisdom that such distance parameters should increase slowly as a function of the number of nodes (like O(log n) or O(log(log n)). Existing graph generation models do not exhibit these types of behavior even at a qualitative level. We provide a new graph generator, based on a forest fire spreading process that has a simple, intuitive justification, requires very few parameters (like the flammability of nodes), and produces graphs exhibiting the full range of properties observed both in prior work and in the present study. We also notice that the forest fire model exhibits a sharp transition between sparse graphs and graphs that are densifying. Graphs with decreasing distance between the nodes are generated around this transition point. Last, we analyze the connection between the temporal evolution of the degree distribution and densification of a graph. We find that the two are fundamentally related. We also observe that real networks exhibit this type of relation between densification and the degree distribution.

Leskovec et al.: Graphs over time: densification laws, shrinking diameters and possible explanations leskovec2005graphs

Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. "Graphs over time: densification laws, shrinking diameters and possible explanations". In: *Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining*. KDD05. ACM, Aug. 2005, pp. 177–187. DOI: 10.1145/1081870.1081893.

Annotations: **05/30/26**: The two counter-intuitive trends are 1. average out-degree increases over time, and 2. effective diameter decreases over time, both following power-laws. The first might make sense: networks have a missing past and they accumulate more preference as they grow (but this depends on the network type). More precisely, they propose a plausible model that has these trends, dubbed Community Guided Attachment, who cite according to a power-law function (scale-free) of the inter-community distance (i.e. the depth of the least-common-ancestor in a hierarchical community). The second forest-fire model further explains the shrinking effective diameter. A node links to its generator, selecting nodes (geometrically) in its 1-hop to continue burning. I think this is a good example of how to link theory to experiment.

Leskovec et al.: Empirical comparison of algorithms for network community detection leskovec2010empirical

Jure Leskovec, Kevin J. Lang, and Michael Mahoney. "Empirical comparison of algorithms for network community detection". In: *Proceedings of the 19th international conference on World wide web*. WWW '10. ACM, Apr. 2010, pp. 631–640. DOI: 10.1145/1772690.1772755.

Annotations: **05/21/26**: This paper is closely related to the evaluation in Yang and Leskovec [42]. Using the same objective functions, they evaluate various clustering algorithms on. This paper also considers size resolved cluster quality, i.e. the NCP, typically a V shape on log-log scales. This paper gives another set of examples for *discriminative* qualities (bias towards certain types of clusters), and inductive bias of algorithms (e.g. separability vs compactness, even for same objective). This paper is okay; its a good step towards bridging theory and practice, but applied to flow/spectral algorithms which I don't know much about.

Abstract: Detecting clusters or communities in large real-world graphs such as large social or information networks is a problem of considerable interest. In practice, one typically chooses an objective function that captures the intuition of a network cluster as set of nodes with better internal connectivity than external connectivity, and then one applies approximation algorithms or heuristics to extract sets of nodes that are related to the objective function and that "look like" good communities for the application of interest. In this paper, we explore a range of network community detection methods in order to compare them and to understand their relative performance and the systematic biases in the clusters they identify. We evaluate several common objective functions that are used to formalize the notion of a network community, and we examine several different classes of approximation algorithms that aim to optimize such objective functions. In addition, rather than simply fixing an objective and asking for an approximation to the best cluster of any size, we consider a size-resolved version of the optimization problem. Considering community quality as a function of its size provides a much finer lens with which to examine community detection algorithms, since objective functions and approximation algorithms often have non-obvious size-dependent behavior.

Leskovec et al.: Community Structure in Large Networks: Natural Cluster Sizes and the Absence of Large Well-Defined Clusters leskovec2009community

Jure Leskovec et al. "Community Structure in Large Networks: Natural Cluster Sizes and the Absence of Large Well-Defined Clusters". In: *Internet Mathematics* 6.1 (Jan. 2009), pp. 29–123. ISSN: 1944-9488. DOI: 10.1080/15427951.2009.10129177.

Abstract: A large body of work has been devoted to defining and identifying clusters or communities in social and information networks, i.e., in graphs in which the nodes represent underlying social entities and the edges represent some sort of interaction between pairs of nodes. Most such research begins with the premise that a community or a cluster should be thought of as a set of nodes that has more and/or better connections between its members than to the remainder of the network. In this paper, we explore from a novel perspective several questions related to identifying meaningful communities in large social and information networks, and we come to several striking conclusions. Rather than defining a procedure to extract sets of nodes from a graph and then attempting to interpret these sets as "real" communities, we employ approximation algorithms for the graph-partitioning problem to characterize as a function of

size the statistical and structural properties of partitions of graphs that could plausibly be interpreted as communities. In particular, we define the network community profile plot, which characterizes the "best" possible community–according to the conductance measure–over a wide range of size scales. We study over one hundred large real-world networks, ranging from traditional and online social networks, to technological and information networks and web graphs, and ranging in size from thousands up to tens of millions of nodes. Our results suggest a significantly more refined picture of community structure in large networks than has been appreciated previously. Our observations agree with previous work on small networks, but we show that large networks have a very different structure. In particular, we observe tight communities that are barely connected to the rest of the network at very small size scales (up to $\approx$ 100 nodes); and communities of size scale beyond $\approx$ 100 nodes gradually "blend into" the expander-like core of the network and thus become less "community-like," with a roughly inverse relationship between community size and optimal community quality. This observation agrees well with the so-called Dunbar number, which gives a limit to the size of a well-functioning community. However, this behavior is not explained, even at a qualitative level, by any of the commonly used network-generation models. Moreover, it is exactly the opposite of what one would expect based on intuition from expander graphs, low-dimensional or manifold-like graphs, and from small social networks that have served as test beds of community-detection algorithms. The relatively gradual increase of the network community profile plot as a function of increasing community size depends in a subtle manner on the way in which local clustering information is propagated from smaller to larger size scales in the network. We have found that a generative graph model, in which new edges are added via an iterative "forest fire" burning process, is able to produce graphs exhibiting a network community profile plot similar to what we observe in our network data sets.

Mahmoudi et al.: Proof of biased behavior of Normalized Mutual Information mahmoudi2024proof

Amin Mahmoudi and Dariusz Jemielniak. "Proof of biased behavior of Normalized Mutual Information". In: *Scientific Reports* 14.1 (Apr. 2024). ISSN: 2045-2322. DOI: `10.1038/s41598-024-59073-9`.

Annotations: **05/30/26**: This seems like its written by higher-schoolers for high schoolers. Proving immediate properties of NMI and no clear theorem. Figure 1 is fine.

Abstract: The Normalized Mutual Information (NMI) metric is widely utilized in the evaluation of clustering and community detection algorithms. This study explores the performance of NMI, specifically examining its performance in relation to the quantity of communities, and uncovers a significant drawback associated with the metric's behavior as the number of communities increases. Our findings reveal a pronounced bias in the NMI as the number of communities escalates. While previous studies have noted this biased behavior, they have not provided a formal proof and have not addressed the causation of this problem, leaving a gap in the existing literature. In this study, we fill this gap by employing a mathematical approach to formally demonstrate why NMI exhibits biased behavior, thereby establishing its unsuitability as a metric for evaluating clustering and community detection algorithms. Crucially, our study exposes the vulnerability of entropy-based metrics that employ logarithmic functions to similar bias.

Malvestio et al.: Interplay between k-core and community structure in complex networks malvestio2020interplay

Irene Malvestio, Alessio Cardillo, and Naoki Masuda. "Interplay between k-core and community structure in complex networks". In: *Scientific Reports* 10.1 (Sept. 2020). ISSN: 2045-2322. DOI: `10.1038/s41598-020-71426-8`.

Annotations: **05/26/26**: Finds that matching degree sequence (i.e. configuration model) is not sufficient to reconstruct k-core distribution (by survival curve $1 - CDF(k_{shell})$). LFR networks are worse than SBM synthetic at reconstructing coreness, but neither seem all too good. Good paper, and results reinforce our understanding that SBM and LFR generators are bad; how would RECCS/EC-SBM do?

Abstract: The organisation of a network in a maximal set of nodes having at least k neighbours within the set, known as k-core decomposition, has been used for studying various phenomena. It has been shown that nodes in the innermost k-shells play a crucial role in contagion processes, emergence of consensus, and resilience of the system. It is known that the k-core decomposition of many empirical networks cannot be explained by the degree of each node alone, or equivalently, random graph models that preserve the degree of each node (i.e., configuration model). Here we study the k-core decomposition of some empirical networks as well as that of some randomised counterparts, and examine the extent to which the k-shell structure of the networks can be accounted for by the community structure. We find that preserving the community structure in the randomisation process is crucial for generating networks whose k-core decomposition is close to the empirical one. We also highlight the existence, in some networks, of a concentration of the nodes in the innermost k-shells into a small number of communities.

Vinh et al.: Information theoretic measures for clusterings comparison: is a correction for chance necessary? vinh2009information

Nguyen Xuan Vinh, Julien Epps, and James Bailey. "Information theoretic measures for clusterings comparison: is a correction for chance necessary?" In: *Proceedings of the 26th Annual International Conference on Machine Learning*. ICML '09. ACM, June 2009, pp. 1073–1080. DOI: `10.1145/1553374.1553511`.

Annotations: **05/30/26**: This is the original AMI (adjusted mutual information) paper. It gives two nice experiments for why NMI should be corrected, i.e. with the average score of a permutationally random clustering with K clusters, and the expected pairwise difference between two clusterings of size K. Straightforward derivation using the hypergeometric distribution for the expectation, and discusses nice ways to adjust using it. Can black-box, and should (in order for reasonable publishable results).

Abstract: Information theoretic based measures form a fundamental class of similarity measures for comparing clusterings, beside the class of pair-counting based and set-matching based measures. In this paper, we discuss the necessity of correction for chance for information theoretic based measures for clusterings comparison. We observe that the baseline for such measures, i.e. average value between random partitions of a data set, does not take on a constant value, and tends to have larger variation when the ratio between the number of data points and the number of clusters is small. This effect is similar in some other non-information theoretic based measures such as the well-known Rand Index. Assuming a hypergeometric model of randomness, we derive the analytical formula for the expected mutual information value between a pair of clusterings, and then propose the adjusted version for several popular information theoretic based measures. Some examples are given to demonstrate the need and usefulness of the adjusted measures.

Yang et al.: Defining and evaluating network communities based on ground-truth yang2012defining

Jaewon Yang and Jure Leskovec. "Defining and evaluating network communities based on ground-truth". In: *Proceedings of the ACM SIGKDD Workshop on Mining Data Semantics*. KDD '12. ACM, Aug. 2012, pp. 1–8. DOI: `10.1145/2350190.2350193`.

Annotations: **05/21/26**: This paper allows for overlapping clusters, as they are scored individually, and split by components. Distinction between goodness and scoring is still a bit confusing – whats wrong with using goodness as a scoring criterion? Also the cohesiveness formula is a bit strange, but similar to NCP, and perhaps densest subgraph. The clustering coefficient is defined as $g(S) = \frac{1}{|S|}\sum_{v\in S}\frac{2|E(N(v))|}{|N(v)|\cdot(|N(v)|-1)}$, it and density are similar to tpr. Cohesiveness is defined as maximum subgraph conductance, it and separability are similar to conductance. Desirable traits are stable under small perturbations, but sharp increase with large perturbation — evaluation familiar to Fisher information (more sensitive is more information). 6 distinct sources of networks (225/230 of them are from same platform). *Thought experiment: suppose clustering is inaccurate. What does this paper tell us?* We don't know the node coverage. **05/18/26**: There are three barriers for evaluating accuracy of clustering algorithms on empirical: 1. lack of datasets, 2. lack of reliable ground-truth, and 3. a common definition for what a community is. The authors propose 4 axioms (community goodness) that a good community is, namely it must be separable, dense, cohesive, and have high clustering coefficient. They evaluate 13 scoring functions, finding that modularity tends to be negatively correlated with a good community and suggest joint evaluation using conductance and TPR for robust evaluation. Finally, using these functions, they procure a set of datasets with (somewhat) reliable ground-truth. This paper is OK; takeaway is their procedure of evaluating scoring methods, and the ridiculing of modularity, but the ground-truth they create from them is questionable.

Abstract: Nodes in real-world networks, such as social, information or technological networks, organize into communities where edges appear with high concentration among the members of the community. Identifying communities in networks has proven to be a challenging task mainly due to a plethora of definitions of a community, intractability of algorithms, issues with evaluation and the lack of a reliable gold-standard ground-truth. We study a set of 230 large social, collaboration and information networks where nodes explicitly define group memberships. We use these groups to define the notion of ground-truth communities. We then propose a methodology which allows us to compare and quantitatively evaluate different definitions of network communities on a large scale. We choose 13 commonly used definitions of network communities and examine their quality, sensitivity and robustness. We show that the 13 definitions naturally group into four classes. We find that two of these definitions, Conductance and Triad-participation-ratio, consistently give the best performance in identifying ground-truth communities.

Marked for Re-Reading

Gou et al.: A Comprehensive Survey and Experimental Study of Learning-Based Community Search gou2025comprehensive

Xiaoxuan Gou et al. "A Comprehensive Survey and Experimental Study of Learning-Based Community Search". In: *Proceedings of the VLDB Endowment* 18.9 (May 2025), pp. 2941–2954. ISSN: 2150-8097. DOI: 10.14778/3746405.3746419.

Abstract: Given a graph G and a query node q, the goal of community search (CS) is to find a structurally cohesive subgraph from G that contains q. Significant progress has been made in community search using deep learning in recent years. To the best of our knowledge, no existing work has provided a comprehensive review of learning-based community search methods. Additionally, we find that: (1) Existing methods offer diverse definitions or descriptions of communities, which require systematic summarization. (2) The methods rely on distinct metrics for limited community assessment. (3) Overhead evaluations of the methods vary and exhibit certain biases. Therefore, a comprehensive survey and experimental study are essential to achieve four key objectives: designing a unified pipeline, clarifying community definitions, enriching community evaluation, and establishing overhead assessment. In this paper, we first propose a unified pipeline for these methods, highlighting techniques. We categorize community definitions and analyze the relationships between identified communities. Beyond that, we proposed several community metrics to evaluate the communities comprehensively. Moreover, we introduce a more detailed overhead evaluation approach that considers resource consumption during both the training and search phases. Finally, we employ the proposed community evaluation metrics and overhead assessment framework to evaluate and analyze the methods, examine correlations among metrics, and explore the effects of several commonly used techniques.

Gou et al.: Effective and Efficient Community Search with Graph Embeddings gou2023effective

Xiaoxuan Gou et al. "Effective and Efficient Community Search with Graph Embeddings". In: *ECAI 2023*. IOS Press, Sept. 2023. ISBN: 9781643684376. DOI: 10.3233/faia230358.

Annotations: **05/21/26**: k-core based community search (query sets) via graph convolutional network; requires training; I hate machine learning papers. Sometimes okay, but also cases there get $<$ 50% F1 score with true solution. maximum dataset network size is 60,000.

Abstract: Given a graph G and a query node q , community search (CS) seeks a cohesive subgraph from G that contains q . CS has gained much research interests recently. In the database research community, researchers aim to find the most cohesive subgraph satisfying a specific community model (e.g., k -core or k -truss) via graph traversal. These works obtain good precision, however suffering from the low efficiency issue. In the AI research community, a new thought of using the deep learning model to support CS without relying on graph traversal emerges. Supervised end-to-end models using GCN are presented, which perform efficiently, but leave a large room for precision improvement. None of them can achieve a good balance between the efficiency and effectiveness. This motivates our solution: First, we present an offline community-injected graph embedding method to preserve the community's cohesiveness features into the learned node representations. Second, we resort to a proximity graph (PG) built from node representations, to quickly return the community online. Moreover, we develop a self-augmented method based on KL divergence to further optimize node representations. Extensive experiments on seven real-world graphs show our solution's superiority on effectiveness (at least 39.3% improvement) and efficiency (one to two orders of magnitude faster).

Jerdee et al.: Normalized mutual information is a biased measure for classification and community detection jerdee2025normalized

Maximilian Jerdee, Alec Kirkley, and Mark Newman. "Normalized mutual information is a biased measure for classification and community detection". In: *Nature Communications* 16.1 (Dec. 2025). ISSN: 2041-1723. DOI: 10.1038/s41467-025-66150-8.

Annotations: **05/24/26**: This is yet another way of improving NMI (e.g. AMI and RMI). I don't think they compare it to AMI or RMI. NMI is size-biased (prefers many estimated groups), and the normalization factor should be different for different clusterings (asymmetric). Working through the math may be instructive but not easy. They only compare to Infomap and modularity optimization, neither are known for their accuracy.

Abstract: Normalized mutual information is widely used as a similarity measure for evaluating the performance of clustering and classification algorithms. In this paper, we argue that results returned by the normalized mutual information are biased for two reasons: first, because they ignore the information content of the contingency table and, second, because their symmetric normalization introduces spurious dependence on algorithm output. We introduce a modified version of the mutual information that remedies both of these shortcomings. As a practical demonstration of the importance of using an unbiased measure, we perform extensive numerical tests on a basket of popular algorithms for network community detection and show that one's conclusions about which algorithm is best are significantly affected by the biases in the traditional mutual information. From blood tests to friend groups, normalized mutual information is widely used to assess similarity between classifications, outcomes, or labelings of data. Here the authors demonstrate systematic biases of this measure and propose a modification that eliminates them.

Sozio et al.: The community-search problem and how to plan a successful cocktail party sozio2010communitysearch

Mauro Sozio and Aristides Gionis. "The community-search problem and how to plan a successful cocktail party". In: *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*. KDD '10. ACM, July 2010, pp. 939–948. DOI: 10.1145/1835804.1835923.

Annotations: **05/21/26**: This is the original paper for community search. The problem is to find the maximium score, connected (not necessarily the largest size community). They also often add a distance constraint. Describes the Local and Global search problem, and an exact greedy algorithm that optimizes a small family of objectives.

Abstract: A lot of research in graph mining has been devoted in the dis-covery of communities. Most of the work has focused in thescenario where communities need to be discovered with onlyreference to the input graph. However, for many interest-ing applications one is interested in finding the communityformed by a given set of nodes. In this paper we study aquery-dependent variant of the community-detection prob-lem, which we call the community-search problem: given agraph G, and a set of query nodes in the graph, we seek to find a subgraph of G that contains the query nodes and itis densely connected.We motivate a measure of density based on minimum de-gree and distance constraints, and we develop an optimumgreedy algorithm for this measure. We proceed by charac-terizing a class of monotone constraints and we generalizeour algorithm to compute optimum solutions satisfying anyset of monotone constraints. Finally we modify the greedyalgorithm and we present two heuristic algorithms that find communities of size no greater than a specified upper bound.Our experimental evaluation on real datasets demonstratesthe efficiency of the proposed algorithms and the quality ofthe solutions we obtain.

Tabatabaee et al.: FastEnsemble: Scalable ensemble clustering on large networks tabatabaee2025fastensemble

Yasamin Tabatabaee et al. "FastEnsemble: Scalable ensemble clustering on large networks". In: *PLOS Complex Systems* 2.10 (Oct. 2025). Ed. by Tobias Braun, e0000069. ISSN: 2837-8830. DOI: 10.1371/journal.pcsy.0000069.

Abstract: Many community detection algorithms are inherently stochastic, leading to variations in their output depending on input parameters and random seeds. This variability makes the results of a single run of these algorithms less reliable. Moreover, different clustering algorithms, optimization criteria (e.g., modularity and the Constant Potts model), and resolution values can result in substantially different partitions on the same network. Consensus clustering methods, such as Ensemble Clustering for Graphs (ECG) and FastConsensus, have been proposed to reduce the instability of non-deterministic algorithms and improve their accuracy by combining a set of partitions resulting from multiple runs of a clustering algorithm. In Complex Networks and their Applications 2024, we introduced FastEnsemble, a new consensus clustering method; here we present a more extensive evaluation of this method. Our results on both real-world and synthetic networks show that FastEnsemble produces more accurate clusterings than two other consensus clustering methods, ECG and FastConsensus, for many model conditions. Furthermore, FastEnsemble is fast enough to be used on networks with more than 3 million nodes, and so improves on the speed and scalability of FastConsensus. Finally, we showcase the utility of consensus clustering methods in mitigating the effect of resolution limit and clustering networks that are only partially covered by communities.

Tang et al.: Reliable community search in dynamic networks **tang2022reliable**

Yifu Tang et al. "Reliable community search in dynamic networks". In: *Proceedings of the VLDB Endowment* 15.11 (July 2022), pp. 2826–2838. ISSN: 2150-8097. DOI: `10.14778/3551793.3551834`.

Abstract: Searching for local communities is an important research problem that supports advanced data analysis in various complex networks, such as social networks, collaboration networks, cellular networks, etc. The evolution of such networks over time has motivated several recent studies to identify local communities in dynamic networks. However, these studies only utilize the aggregation of disjoint structural information to measure the quality and ignore the reliability of the communities in a continuous time interval. To fill this research gap, we propose a novel (θ, k)-core reliable community (CRC) model in the weighted dynamic networks, and define the problem of most reliable community search that couples the desirable properties of connection strength, cohesive structure continuity, and the maximal member engagement. To solve this problem, we first develop a novel edge filtering based online CRC search algorithm that can effectively filter out the trivial edge information from the networks while searching for a reliable community. Further, we propose an index structure, Weighted Core Forest-Index (WCF-index), and devise an index-based dynamic programming CRC search algorithm, that can prune a large number of insignificant intermediate results and support efficient query processing. Finally, we conduct extensive experiments systematically to demonstrate the efficiency and effectiveness of our proposed algorithms on eight real datasets under various experimental settings.

Watts et al.: Collective dynamics of 'small-world' networks **watts1998collective**

Duncan J. Watts and Steven H. Strogatz. "Collective dynamics of 'small-world' networks". In: *Nature* 393.6684 (June 1998), pp. 440–442. ISSN: 1476-4687. DOI: `10.1038/30918`.

Annotations: **05/21/26**: This short article defines the clustering coefficient (average over vertices of edge density in neighborhoods) and the characteristic time (average of APSP matrix). This invents the notion of small-world networks and proposes a (parameterized) random process that can describe the behavior of these stats. They use this model to investigate functional traits, such as disease spread. Apparently this model is very different from previous models – second look would be good.

Abstract: Networks of coupled dynamical systems have been used to model biological oscillators, Josephson junction arrays, excitable media, neural networks, spatial games, genetic control networks and many other self-organizing systems. Ordinarily, the connection topology is assumed to be either completely regular or completely random. But many biological, technological and social networks lie somewhere between these two extremes. Here we explore simple models of networks that can be tuned through this middle ground: regular networks 'rewired' to introduce increasing amounts of disorder. We find that these systems can be highly clustered, like regular lattices, yet have small characteristic path lengths, like random graphs. We call them 'small-world' networks, by analogy with the small-world phenomenon (popularly known as six degrees of separation). The neural network of the worm Caenorhabditis elegans, the power grid of the western

United States, and the collaboration graph of film actors are shown to be small-world networks. Models of dynamical systems with small-world coupling display enhanced signal-propagation speed, computational power, and synchronizability. In particular, infectious diseases spread more easily in small-world networks than in regular lattices.

Miscellaneous

Castillo-Castillo et al.: A growth model for citations networks **castillo-castillo2025growth**

Pedro Castillo-Castillo et al. "A growth model for citations networks". In: *Applied Network Science* 10.1 (Jan. 2025). ISSN: 2364-8228. DOI: `10.1007/s41109-025-00691-1`.

Annotations: **06/01/26**: First, they provide a nice comparison of many previous preferential-attachment network generations, such as the original BA (degree centrality), Yule model (each node doubles, new communities form if sufficiently distinct), and GNR model (where we have hierarchical nodes (DAG) and we cite upwards with higher probability). The authors synthesize these ideas for citations (community sizes follow chinese-restaurant process), showing several nice patterns such as power-law in-degree/out-degree/community-size. Meh paper – their discussion is a bit weak and they seem to suggest it as a plausible model for real networks without sufficient evidence.

Abstract: In some complex networks, nodes can continuously increase or decrease the number of their incoming and outgoing links. The World Wide Web is one such example, as webmasters can add or delete hyperlinks on web pages at any time. There are also networks where this can not happen. For instance, in citation networks of scientific articles, after an article has been published, it will start gaining incoming links as it is cited, but its outgoing links will remain unchanged. Although articles are published with a predetermined number of references, the distribution of their outgoing links follows a power law, as if they were the result of a preferential process. So, how can we explain that the number of references an author includes in a scientific article is not purely random? In this work, a growth model for citation networks is presented, proposing that the distribution of outgoing links can be shaped by the presence of communities.

Chacko et al.: An agent-based model of citation behavior **chacko2026agentbased**

George Chacko et al. "An agent-based model of citation behavior". In: *Applied Network Science* 11.1 (Jan. 2026). ISSN: 2364-8228. DOI: `10.1007/s41109-025-00766-z`.

Annotations: **05/29/26**: This paper describes a model for generating citation networks, with $u \to v$ edge representing the article u citing article v. Similar to Leskovec, Kleinberg, and Faloutsos [35], their model has agents who cite within the neighborhood of its generator (but with rate α). What differs is that each paper has a vertex weight, based upon recency, fitness, and preferential attachment. The main efforts are towards scaling and testing hypothesis of citation behavior, but this paper only does some rudimentary analysis, such as sensitivity of the model and the extent to which tuning up each parameter affects the in-degree of a node.

Abstract: Whether citations objectively and reliably reflect the quality of articles and researchers is questionable. Even so, citation counts are widely used to estimate the productivity of researchers and institutions, which creates a 'grubby' motivation to be well-cited. We examine this motivation using a generative model of citation that is agent-based. In this model, new nodes are added to an existing citation network. These new nodes act as autonomous agents that cite other nodes based on a composite bias for preferential attachment, recency, fitness (epistemic quality), and community structure. We use the model to ask whether strategic citation behaviors can support an interest in being well-cited. Results from this model suggest that while fitness is influential, the number of references and community effects are also influential in attracting citations. These results raise questions about similar effects in the real world.

Clauset et al.: Power-Law Distributions in Empirical Data **clauset2009powerlaw**

Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. "Power-Law Distributions in Empirical Data". In: *SIAM Review* 51.4 (Nov. 2009), pp. 661–703. ISSN: 1095-7200. DOI: `10.1137/070710111`.

Annotations: **05/29/26**: For our purposes, Box 1 describes the procedure for analyzing power-law distributed data and the rest of the paper is not too important. Important note is that we are analyzing *tails*, rather than the entire distribution (compare to exponential tail in ER graph for example).

Abstract: Power-law distributions occur in many situations of scientific interest and have significant consequences for our understanding of natural and man-made phenomena. Unfortunately, the detection and characterization of power laws is complicated by the large fluctuations that occur in the tail of the distribution–the part of the distribution representing large but rare events–and by the difficulty of identifying the range over which power-law behavior holds. Commonly used methods for analyzing power-law data, such as least-squares fitting, can produce substantially inaccurate estimates of parameters for power-law distributions, and even in cases where such methods return accurate answers they are still unsatisfactory because they give no indication of whether the data obey a power law at all. Here we present a principled statistical framework for discerning and quantifying power-law behavior in empirical data. Our approach combines maximum-likelihood fitting methods with goodness-of-fit tests based on the Kolmogorov-Smirnov (KS) statistic and likelihood ratios. We evaluate the effectiveness of the approach with tests on synthetic data and give critical comparisons to previous approaches. We also apply the proposed methods to twenty-four real-world data sets from a range of different disciplines, each of which has been conjectured to follow a power-law distribution. In some cases we find these conjectures to be consistent with the data, while in others the power law is ruled out.

Dhulipala et al.: Theoretically Efficient Parallel Graph Algorithms Can Be Fast and Scalable dhulipala2021theoretically

Laxman Dhulipala, Guy E. Blelloch, and Julian Shun. "Theoretically Efficient Parallel Graph Algorithms Can Be Fast and Scalable". In: *ACM Transactions on Parallel Computing* 8.1 (Mar. 2021), pp. 1–70. ISSN: 2329-4957. DOI: 10.1145/3434393.

Annotations: **05/24/26**: Presents the graph bencharmks and has work-efficient but also high performing shared-memory implementations. Showcases what a good parallel framework can do. Really impressive results, such as k-core decomposition in seconds on billion-node graphs (72 CPU cores). Presents descriptions and implementations of SOTA parallel graph algos.

Kamiński et al.: Mining complex networks kaminski2021mining

Bogumił Kamiński, Paweł Prałat, et al. *Mining complex networks*. Chapman and Hall/CRC, 2021.

Lü et al.: The H-index of a network node and its relation to degree and coreness lu2016hindex

Linyuan Lü et al. "The H-index of a network node and its relation to degree and coreness". In: *Nature Communications* 7.1 (Jan. 2016). ISSN: 2041-1723. DOI: 10.1038/ncomms10168.

Abstract: Identifying influential nodes in dynamical processes is crucial in understanding network structure and function. Degree, H-index and coreness are widely used metrics, but previously treated as unrelated. Here we show their relation by constructing an operator , in terms of which degree, H-index and coreness are the initial, intermediate and steady states of the sequences, respectively. We obtain a family of H-indices that can be used to measure a node's importance. We also prove that the convergence to coreness can be guaranteed even under an asynchronous updating process, allowing a decentralized local method of calculating a node's coreness in large-scale evolving networks. Numerical analyses of the susceptible-infected-removed spreading dynamics on disparate real networks suggest that the H-index is a good tradeoff that in many cases can better quantify node influence than either degree or coreness. Identifying influential nodes in networks is important for the understanding of their structure and function, but there are several so far unrelated measures to assess this. Here, the authors unfold relations among knows criteria and construct a family of indices that interpolate between degree and coreness.

Newman: Mixing patterns in networks — newman2003mixing

M. E. J. Newman. "Mixing patterns in networks". In: *Physical Review E* 67.2 (Feb. 2003). ISSN: 1095-3787. DOI: 10.1103/physreve.67.026126.

Annotations: **06/12/26**: Assortativity is the tendancy for like-to-like connections (discrete). Studying assortativity naturally leads to formation of communities, as well as characterizing important network structures such as viral spread and network resilience. The main tool is the assortativity coefficient; assign each node a discrete scalar quantity, and let e_{ij} be fraction of edges between cartegory i and j. Then, r is the deviation from $\sum_i e_{ii}$ from its null counterpart (i.e. w/ no mixing, using fractional edges), normalized to be within -1 and 1. They study this quantity for many types of networks and demonstrate the relationship to other network characteristics (above). One notable example is that social networks have high assortativity, leading to an increased vulnerability to viral spread compared to the internet, which is more dissortative. Somewhat cool paper – heavy discussion on the models which I haven't (and don't want to) read in depth.

Abstract: We study assortative mixing in networks, the tendency for vertices in networks to be connected to other vertices that are like (or unlike) them in some way. We consider mixing according to discrete characteristics such as language or race in social networks and scalar characteristics such as age. As a special example of the latter we consider mixing according to vertex degree, i.e., according to the number of connections vertices have to other vertices: do gregarious people tend to associate with other gregarious people? We propose a number of measures of assortative mixing appropriate to the various mixing types, and apply them to a variety of real-world networks, showing that assortative mixing is a pervasive phenomenon found in many networks. We also propose several models of assortatively mixed networks, both analytic ones based on generating function methods, and numerical ones based on Monte Carlo graph generation techniques. We use these models to probe the properties of networks as their level of assortativity is varied. In the particular case of mixing by degree, we find strong variation with assortativity in the connectivity of the network and in the resilience of the network to the removal of vertices.

Park et al.: Modeling the Global Citation Network using the Scalable Agent-based Simulator for Citation Analysis with Recency-emphasized Sampling (SASCA-ReS) — park2025modeling

Minhyuk Park et al. *Modeling the Global Citation Network using the Scalable Agent-based Simulator for Citation Analysis with Recency-emphasized Sampling (SASCA-ReS).* en. 2025. DOI: 10.5281/ZENODO.17772113.

Annotations: **05/29/26**: This paper follows up on the work in Chacko et al. [45], both in the model, implementation, and experiments. Citations go to 2-hop, with α the proportion in the 1-hop, and also idiosyncrasies uniformly at random throughout the network. Recency is now a budget instead of a score (i.e. only PA and fitness are scoring, as well as topological constraints). Interesting, predicting whether a node is in the top 0.1% of in-citations can be predicted by a decision tree using out_degree and fitness. They show that α largely affects the clustering coefficients and also the communities (via Leiden-CPM). They also note that mavericks, which are not topologically constrained, typically had much higher disruption and perhaps can artificially inflate their influence.

Wang et al.: Identifying influential nodes through hierarchical k-shell and extended neighborhood integration — wang2026identifying

Feifei Wang et al. "Identifying influential nodes through hierarchical k-shell and extended neighborhood integration". In: *Scientific Reports* 16.1 (Feb. 2026). ISSN: 2045-2322. DOI: 10.1038/s41598-026-40209-y.

Annotations: **05/26/26**: Hierarchical k-shell decomposition means looking at degree (local) and k-shell (global) value, extended neighborhood information means using Jaccard to aggregate influence of one-hop. This is the second paper I've seen discuss influence using the SIR model; first is Lü et al. [49]. The specific formulas make no sense to me.

Abstract: The identification of influential nodes has extensive applications in complex network research. To address the challenge of balancing accuracy and computational efficiency in existing methods, this paper proposes an algorithm named HKEN that integrates hierarchical k-shell decomposition with extended neighborhood information. The approach incorporates both global structural features and local topological information of nodes. First, the coarse-grained hierarchical mechanism of the k-shell algorithm is optimized, and the initial weight of nodes is dynamically computed based on their degree and k-shell values. Second, the neighborhood range of nodes is extended, and a local clustering coefficient is introduced to establish a threshold for information transmission distance. Finally, an influence aggregation strategy based on the Jaccard similarity coefficient is proposed. The performance of HKEN was validated through comparative experiments on 10 real-world networks against 12 benchmark methods. Experimental results demonstrate that the proposed method achieves higher consistency with SIR model outcomes and enhances the propagation capability of top-ranked nodes, confirming its improved identification accuracy.

Watts et al.: Collective dynamics of 'small-world' networks **watts1998collective**

Duncan J. Watts and Steven H. Strogatz. "Collective dynamics of 'small-world' networks". In: *Nature* 393.6684 (June 1998), pp. 440–442. ISSN: 1476-4687. DOI: 10.1038/30918.

Annotations: **05/21/26**: This short article defines the clustering coefficient (average over vertices of edge density in neighborhoods) and the characteristic time (average of APSP matrix). This invents the notion of small-world networks and proposes a (parameterized) random process that can describe the behavior of these stats. They use this model to investigate functional traits, such as disease spread. Apparently this model is very different from previous models – second look would be good.

Abstract: Networks of coupled dynamical systems have been used to model biological oscillators, Josephson junction arrays, excitable media, neural networks, spatial games, genetic control networks and many other self-organizing systems. Ordinarily, the connection topology is assumed to be either completely regular or completely random. But many biological, technological and social networks lie somewhere between these two extremes. Here we explore simple models of networks that can be tuned through this middle ground: regular networks 'rewired' to introduce increasing amounts of disorder. We find that these systems can be highly clustered, like regular lattices, yet have small characteristic path lengths, like random graphs. We call them 'small-world' networks, by analogy with the small-world phenomenon (popularly known as six degrees of separation). The neural network of the worm Caenorhabditis elegans, the power grid of the western United States, and the collaboration graph of film actors are shown to be small-world networks. Models of dynamical systems with small-world coupling display enhanced signal-propagation speed, computational power, and synchronizability. In particular, infectious diseases spread more easily in small-world networks than in regular lattices.

Wernicke: Efficient Detection of Network Motifs **wernicke2006efficient**

Sebastian Wernicke. "Efficient Detection of Network Motifs". In: *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 3.4 (Oct. 2006), pp. 347–359. ISSN: 1545-5963. DOI: 10.1109/tcbb.2006.51.

Annotations: **06/02/26**: A somewhat algorithmic paper on efficient sampling and motifs. Gives a nice way to think about graph isomorphism; seems useful for the triangle counting problem (though more general, I think it is asymptotically optimal). An introduction into the motif significance, but need to read more into it (reminds me a lot about graphons though).

Abstract: Motifs in a given network are small connected subnetworks that occur in significantly higher frequencies than would be expected in random networks. They have recently gathered much attention as a concept to uncover structural design principles of complex networks. Kashtan et al. [Bioinformatics, 2004] proposed a sampling algorithm for performing the computationally challenging task of detecting network motifs. However, among other drawbacks, this algorithm suffers from a sampling bias and scales poorly with increasing subgraph size. Based on a detailed analysis of the previous algorithm, we present a new algorithm for network motif detection which overcomes these drawbacks. Furthermore, we present an efficient new approach for estimating the frequency of subgraphs in random networks that, in contrast to previous approaches, does not require the explicit generation of random networks. Experiments on a testbed of biological networks show our new algorithms to be orders of magnitude faster than previous approaches, allowing for the detection of larger motifs in bigger networks than previously possible and thus facilitating deeper insight into the field.